

Multiple Choice (10 marks)

1. A fair die is tossed 360 times. The probability that a six comes up on 70 or more of the tosses is
 - (a) greater than 0.50
 - (b) between 0.16 and 0.50
 - (c) between 0.02 and 0.16
 - (d) between 0.01 and 0.02
2. Let C be the circle in the yz -plane whose equation is $(y - 3)^2 + z^2 = 1$. If C is revolved around the z -axis, the surface generated is a torus. What is the equation of this torus?
 - (a) $x^2 + y^2 + z^2 + 8 = 6y$
 - (b) $(x^2 + y^2 + z^2)^2 = 8 + 36(x^2 + z^2)$
 - (c) $(x^2 + y^2 + z^2 + 8)^2 = 36(x^2 + z^2)$
 - (d) $(x^2 + y^2 + z^2 + 8)^2 = 36(x^2 + y^2)$
3. Statement 1 | Suppose $f : [a, b]$ is a function and suppose f has a local maximum. $f'(x)$ must exist and equal 0? Statement 2 | There exist non-constant continuous maps from $\mathbb{R}$ to $\mathbb{Q}$.
 - (a) True, True
 - (b) True, False
 - (c) False, True
 - (d) False, False
4. If A is the 2 by 2 matrix whose (i, j) entry is equal to $i + j$, and B is the 3 by 3 matrix whose (i, j) entry is equal to $i + j$, find the value of the sum $\det A + \det B$.
 - (a) -2
 - (b) -1
 - (c) 0
 - (d) 2
5. Statement 1 | $\mathbb{R}$ is a splitting field of some polynomial over $\mathbb{Q}$. Statement 2 | There is a field with 60 elements.
 - (a) True, True
 - (b) False, False
 - (c) True, False
 - (d) False, True

True / False (10 marks)

Answer True or False

6. True or False: There are exactly five positive integers k for which $k!$ ends in exactly 99 zeros.
7. True or False: The probability that at least one integer chosen will be the square of the other is 0.72.
8. True or False: The polynomial $4x - 2$ is irreducible over the integers $\mathbb{Z}$.
9. True or False: The cyclic subgroup of $\mathbb{Z}_{24}$ generated by 18 has an order of 6.
10. True or False: The value of c in $\mathbb{Z}_3$ that makes $\mathbb{Z}_3[x]/(x^3 + x^2 + c)$ a field is 2.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. There is a committee composed of eight women and two men. When they meet, they sit in a row—the women in indistinguishable rocking chairs and the men on indistinguishable stools. How many distinct ways are there for me to arrange the eight chairs and two stools for a meeting?
12. Discuss the implications of the statements that every group of order p^2 (where p is prime) is Abelian and that a Sylow p -subgroup of a group G is normal if and only if it is the only one in G .

End of Paper